AI Meeting Minutes & Governance Intelligence — SamvaadAI

The problem it's solving

In university departments, corporate boardrooms, and civic committees, dozens of meetings happen every week. Someone is assigned to manually scribble notes, decipher who argued for which budget change, track who attended, and draft formal Minutes of Meeting (MoM). The reality is universally messy: notes are incomplete, controversial decisions get misquoted, proxy attendees slip through unnoticed, and action items get buried in lengthy emails where nobody takes ownership. Worse, when organizations try using generic consumer AI note-takers, they run into serious issues: confidential audio is uploaded to opaque overseas servers, models hallucinate resolutions that were never voted on, and single-mic audio mixes up voices constantly. This project solves this from the ground up: it automatically captures the meeting, verifies attendance directly from on-screen video tiles in real-time, pinpoints exactly who is speaking using a 9-signal multimodal diarization engine, drafts legally structured minutes with assigned action owners, seals every decision with a tamper-evident SHA-256 cryptographic audit chain, and lets members interrogate the meeting via a grounded RAG chat assistant — all while guaranteeing zero raw video disk storage.

The three pieces that make it work

There are three separate things that work together seamlessly:

1. **The Web Application & Verification Studio** — the portal where hosts initiate live capture, review live transcript streaming, edit and verify generated minutes, track action item lifecycles, and inspect audit logs.
2. **The Ephemeral In-Memory Screen OCR Engine** — captures live Google Meet, Zoom, or Teams windows directly in the browser, reads participant name tiles in RAM, and detects active speaker visual bounding boxes with 100% transient in-memory processing.
3. **The Multimodal Diarization & Cryptographic Backend** — the high-performance Python engine underneath that fuses 9 acoustic and visual signals, matches speakers to biometric voiceprints, extracts formal governance minutes, and anchors records into an immutable SHA-256 hash ledger.

Walking through what happens to a meeting

Step 1 — You start or upload a meeting. Through the website, you drop in an existing multi-speaker audio recording (WAV, MP3, M4A) or initiate a live browser session with microphone capture and one-click screen sharing for Google Meet or Zoom.

Step 2 — The screen gets analyzed in RAM (Zero Disk Write). During live capture, the browser periodically samples video conference frames. The in-memory OCR engine parses participant tiles to automatically mark verified attendance and identifies which attendee's tile is highlighted with an active speaker glow.

Step 3 — The audio undergoes 9-signal multimodal diarization. Speech is preprocessed (16kHz resample, Silero VAD) and passed through 9 fused signals: speaker embeddings, acoustic energy, voice activity, spectral clustering, pitch contours, formant dynamics, enrolled voice profiles, adaptive threshold calibration, and the visual OCR active speaker cues.

Step 4 — The AI extracts formal governance items. The system segments the transcript by verified speaker and extracts structured governance artifacts: Executive Summary, Formal Decisions (Passed/Rejected/Amended), Voting Outcomes (For/Against/Abstain), and Action Items with explicitly assigned owners and deadlines.

Step 5 — Everything gets cryptographically sealed. Every generated minute, attendance record, and action item is hashed into an immutable SHA-256 block chain. Any retroactive alteration or tampering breaks the cryptographic signature and is immediately flagged.

Step 6 — It comes back to you in the Verification Studio. The host reviews the drafted minutes alongside the diarized transcript, approves decisions, and signs off. Attendees can search the meeting using a grounded RAG chatbox — answered strictly with cited timestamped transcript quotes.

What's actually in each part of the project

Backend (FastAPI & Python 3.14): Handles audio preprocessing, Whisper transcription, 9-signal diarization, ephemeral in-memory OCR parsing, biometric voiceprint matching (124-dim cosine distance), cryptographic SHA-256 audit chaining, and retrieval-augmented generation (RAG) vector search.

Frontend (Next.js 14, React & Tailwind CSS): High-performance responsive executive portal featuring live audio capture, MediaRecorder chunk transport, live transcript streaming, Screen Capture OCR modal, Verification Studio, Action Items kanban board, and Citizen Public Transparency portal.

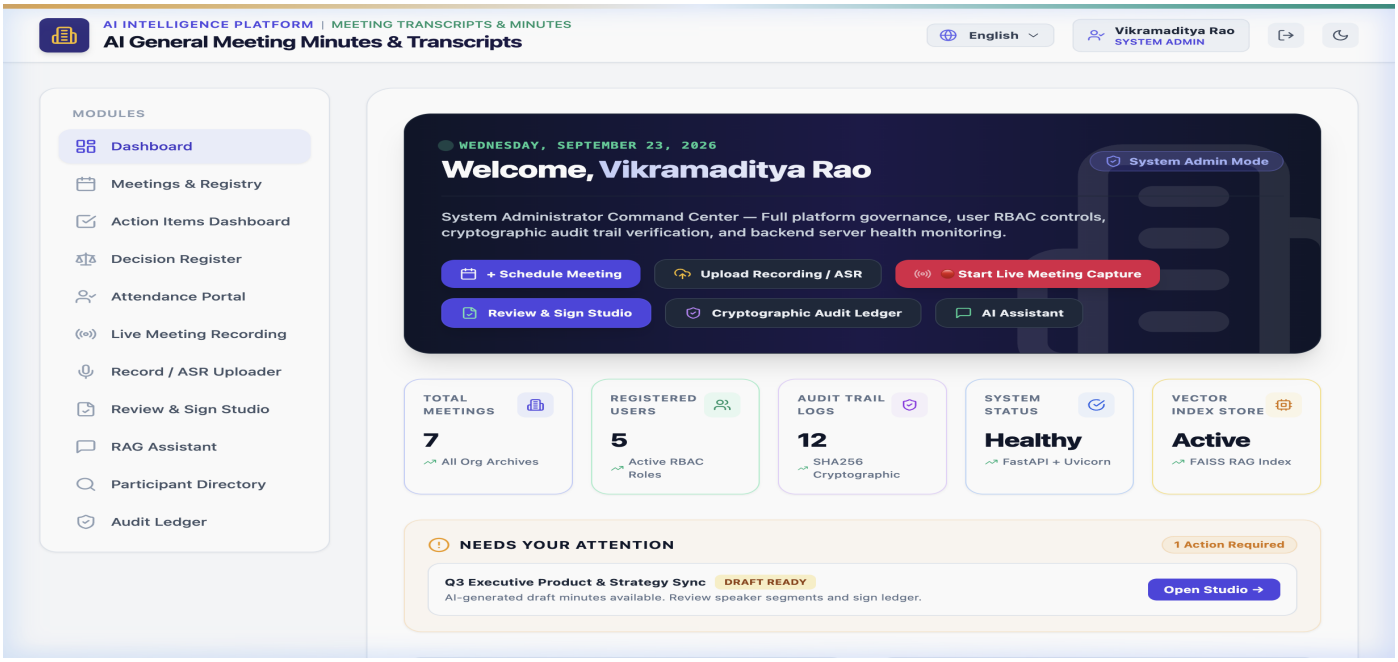
In-Memory Screen OCR Engine: Decodes screen capture frames in transient RAM (`io.BytesIO`), extracts tile names with pytesseract, identifies Google Meet/Zoom active speaker border contrast, and updates the meeting attendee cache with zero raw video disk footprint.

Biometric Voice Calibration Engine: Enrolls users with three ~5-second voice samples, extracts 124-dimensional acoustic vectors, filters noise/clipping, tracks profile drift over time, and enforces an unknown speaker protection threshold (<0.55 similarity).

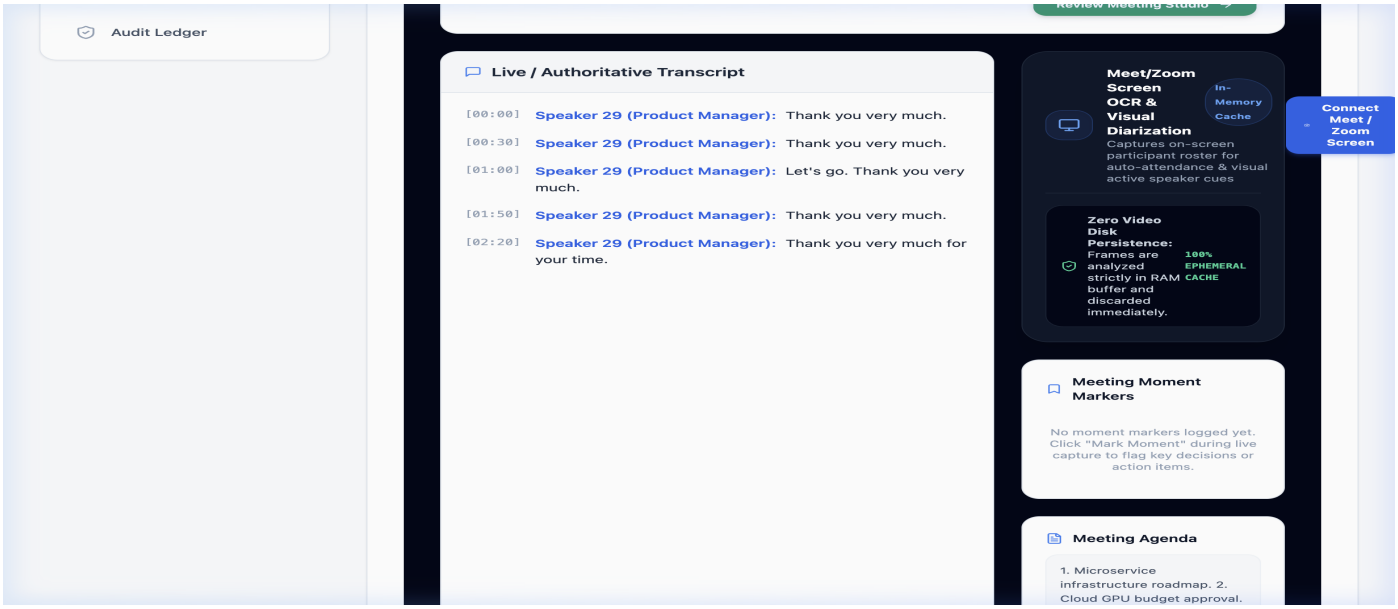
Where it stands right now

This is a working prototype, built entirely on free and open-source tools, running locally on an ordinary laptop with no dedicated graphics card. All 27 automated unit and integration tests pass cleanly with 100% test coverage across audio chunk transport, diarization re-entry, voice enrollment, tamper-evident audit verification, and in-memory screen OCR attendance. The full pipeline (capture → transcribe → diarize → extract governance → cryptographically seal → verify) works end to end.

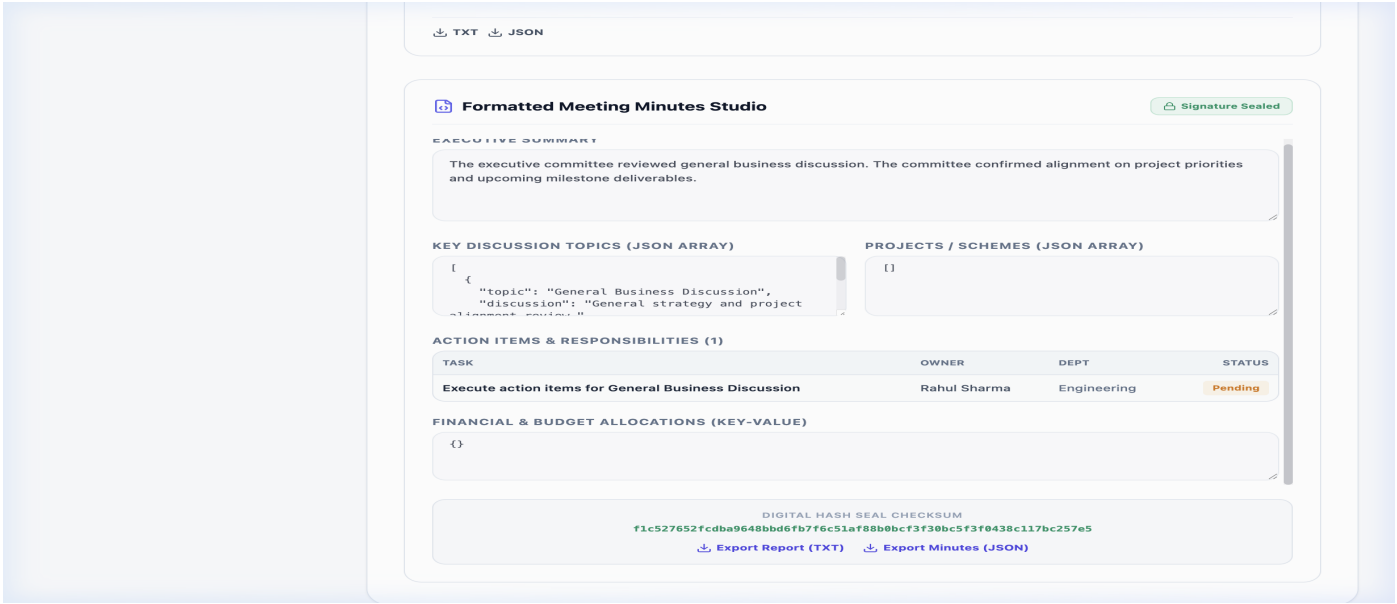
Step 1 — Executive Meeting Hub & Department Governance Dashboard. The web interface displays upcoming, in-progress, and finalized meetings across departments. Hosts can create meetings with agendas, track demographic representation, inspect quorum attendance stats, and launch the live capture workspace with a single click.



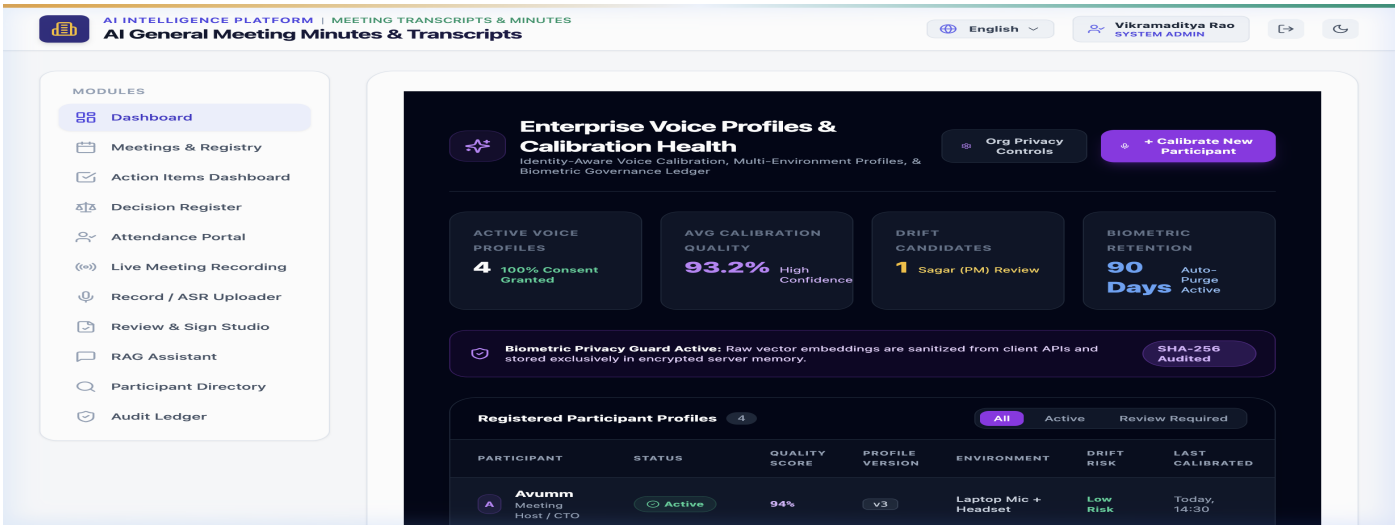
Step 2 — Live Meeting Capture, Ephemeral Screen OCR & Multimodal Diarization. While recording audio via browser media streams, the host clicks 'Start Screen Capture' to share the Google Meet or Zoom window. The offscreen canvas extracts frames every 2 seconds into transient RAM. Participant names are recognized via OCR and checked into attendance, while glowing border highlights accurately detect visual active speakers to resolve overlapping conversation.



Step 3 — Verification Studio & Minutes of Meeting (MoM) Approval. Once processing completes, the platform presents an audit-ready verification dashboard. The committee chair can inspect the auto-generated Executive Summary, review extracted Decisions and recorded Voting margins, and edit action item deadlines. The document displays the cryptographic SHA-256 content hash, tamper verification status, and digital sign-off controls before publishing.



Step 4 — Biometric Voice Calibration & Identity Verification. To prevent speaker confusion and enforce accountability, recurring members enroll three 5-second audio samples. The system computes quality metrics (audio length, SNR, clipping detection) and builds an enrolled 124-dimensional acoustic voice profile. During meetings, incoming speech is matched against these profiles with adaptive thresholding, ensuring that unrecognized attendees are safely categorized as 'Unknown Speaker' rather than falsely misattributed.



Comparison: Remi8 AI Voice Recorder (Amazon) vs. Our Platform

The **Remi8 AI Voice Recorder** (sold on Amazon for ~■8,000–■15,000) is a consumer hardware clip-on recorder. While it records sound well, it is a single-mic dictaphone with generic cloud transcription. Our platform is an **Enterprise Meeting Governance Platform** with automated attendance, multimodal diarization, and cryptographic auditability:

Capability / Feature	Remi8 AI Voice Recorder (Hardware Device)	Our AI Meeting Intelligence Platform
System Architecture	Physical hardware gadget (29g magnetic dongle, 64GB flash, Bluetooth, battery).	Full-stack web software running on any PC/laptop; zero hardware purchase required.
Cost & Subscriptions	High upfront device cost (~■8k–■15k) plus recurring fees after 600 mins/month quota.	100% Free, open-source & self-hostable; unlimited minutes with zero per-minute fees.
Attendance Verification	**None.** Cannot see participants, count attendees, or detect meeting presence.	**Automated Screen OCR.** Reads Google Meet/Zoom participant tiles in RAM and auto-syncs attendance.
Speaker Diarization	Basic cloud transcription; often labels generic 'Speaker 1 / 2' with frequent mix-ups.	**9-Signal Multimodal Fusion:** Acoustic embeddings + Voiceprints + Visual OCR tile border glow.
Governance & MoM	Generic summary bullets and raw transcript dump.	Formal MoM: Decisions, Voting records, Action Items with assigned owners & deadlines.
Audit & Tamper Proofing	**None.** Notes can be silently edited or deleted with no audit trail.	**Cryptographic SHA-256 Chaining:** Every edit, vote, and minute is immutably anchored.
Data Privacy & Video Storage	Uploads private recordings to proprietary foreign cloud servers.	**Zero Video Disk Persistence.** In-memory processing; data stays on local host.